



REDUCING POST HARVEST TOMATO LOSSES IN NIGERIA THROUGH SMART TECHNOLOGY

A hackathon solution built by farmers' logic, not just code

THE PROBLEM

Nigeria grows enough tomatoes to feed itself, yet nearly half never reach a plate

3.8 to 4.0M

metric tonnes of tomatoes
produced in Nigeria every year,
which is the largest output in sub
Saharan Africa

40 to 45%

of that harvest is lost after it
leaves the farm, before it ever
reaches a consumer or processor

Over 50%

is lost during peak harvest season
alone, when market glut and
storage shortages hit hardest

In plain terms, for every 100 baskets a farmer harvests, roughly 40 to 60 of them never earn a single naira. Nigeria still imports large volumes of tomato paste because so much of its own fresh harvest spoils before it can be sold or processed.

ROOT CAUSES

Six weak links that breaks the chain between farm and market

1

Poor transportation

Tomatoes travel long distances on rough roads in overloaded, non refrigerated vehicles, bruises and cracks which speed up spoilage before produce even reaches a market.

2

Inadequate storage

Most smallholders have no access to cold storage. Tomatoes sit at open air temperatures, ripening and decaying within days of harvest.

3

Poor packaging

Woven baskets and sacks offer little protection. The weight of the produce crushes tomatoes at the bottom, and damage spreads to healthy fruit nearby.

4

Market glut and price crashes

During peak harvest, supply outpaces demand, with nowhere to store the surplus, farmers are forced to sell at a loss or watch produce spoil unsold.

5

Pests and disease

Fungal infection, bacterial rot and insect damage spread quickly wherever sanitation, harvesting and handling of product is poor.

6

Weak market coordination

Farmers in remote areas struggle to find buyers quickly. The longer produce sits unsold, the more of it spoils.

OUR OWN PILOT DATA

The numbers behind TomaLink's pilot confirm the scale of the problem

15.75M kg

Total tomatoes
harvested
across our
pilot sample

4M kg

Total quantity
lost before
reaching a
buyer

22.62%

Weighted post
harvest loss
rate across the
pilot

₦10.08bn

Total revenue
generated
from what was
actually sold

₦5.90bn

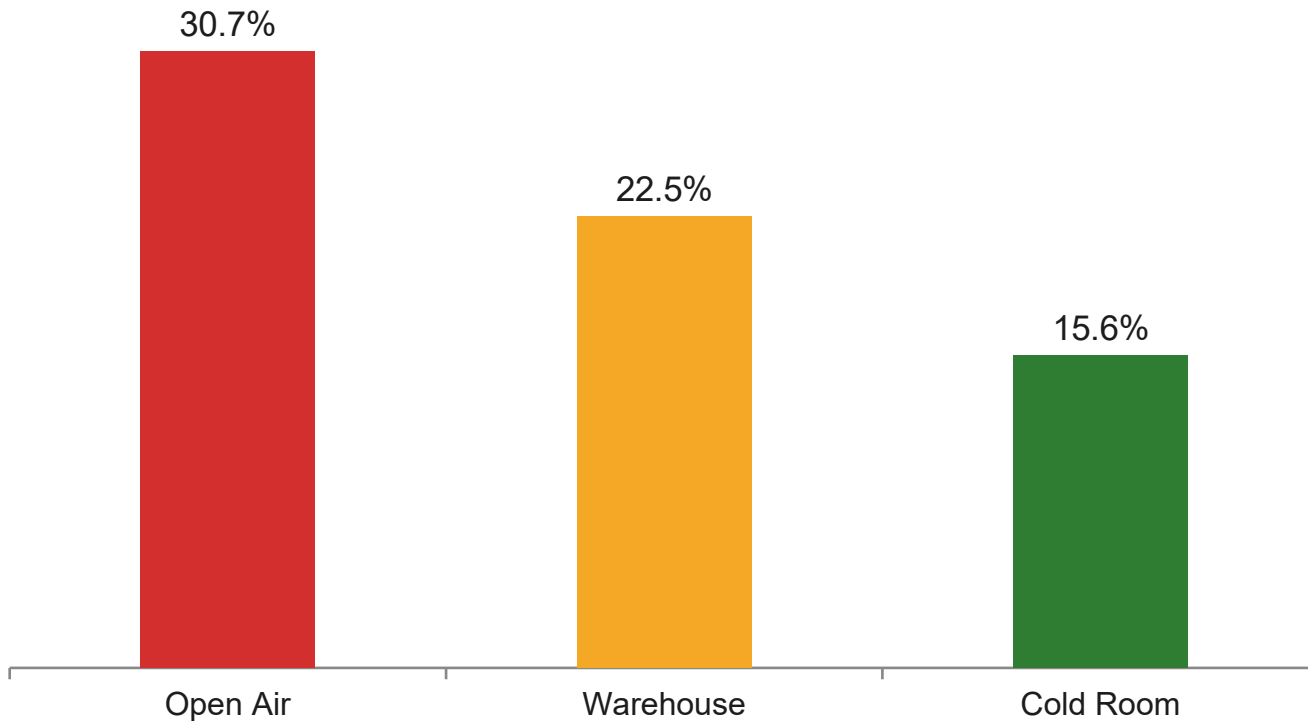
Total profit
realised across
all pilot
farmers

These figures come directly from TomaLink's own pilot loss and impact dashboard, built on real farm level records across five tomato producing states.

WHAT OUR DATA SHOWS

Storage method is the single biggest lever we can pull

Weighted Loss % by Storage Method



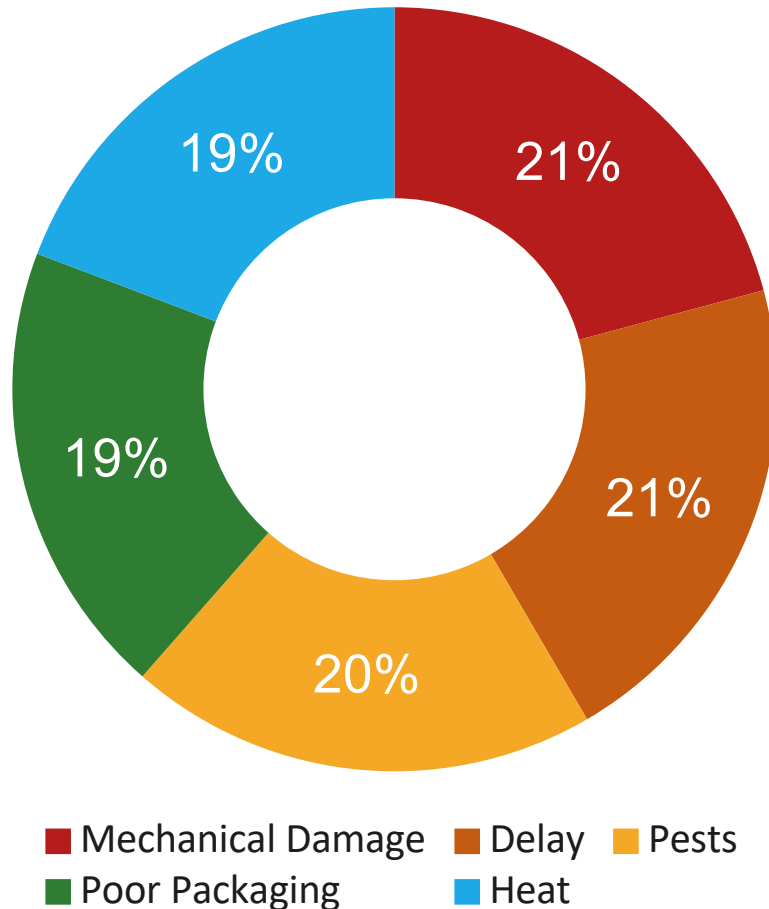
Open air storage loses nearly double the produce that a cold room does.

Every step-up in storage quality buys back real, sellable produce. Moving farmers from open air to even basic warehousing already cuts loss by roughly a third. Cold rooms nearly halve it again.

This is exactly why cold storage and warehouse booking sit at the core of what TomaLink offers, not as an add on feature.

2.0 WHAT OUR DATA SHOWS

How the loss is spread almost evenly across five causes



Mechanical damage and delay top the list, at roughly 20.8% and 20.7% of losses, but pests, packaging and heat all sit close behind.

This spread matters strategically. A platform that only fixes one weak point, only storage, or only transport, would still leave most of the loss untouched. Real impact requires addressing handling, timing, packaging and storage together, in one coordinated flow.

This is the core argument for an integrated platform rather than a single point solution.

WHY THIS MATTERS

Post harvest loss does not stay on the farm, it moves down the whole chain

National Economy

Nigeria loses hundreds of billions of naira annually to tomato spoilage, weakening agriculture's contribution to GDP and forcing continued paste imports that drain foreign exchange.

Consumers

Reduced supply pushes prices up, especially off season, limiting affordable access to a vegetable that is central to Nigerian diets and nutrition.

Farmers

When produce spoils, farmers lose both the harvest and everything already spent on seeds, fertiliser, labour and transport, making it harder to repay loans or reinvest the next season.

Businesses

Processors run below capacity due to inconsistent supply, traders absorb losses on unsold stock, and transporters lose revenue on rejected, damaged produce.

MARKET LANDSCAPE

Every existing platform solves one piece, none solve the whole chain

Side by Side Comparison

Platform	Key Features	Key Advantages	Key Weaknesses
ColdHubs	Solar cold storage in markets, daily crate rental, no grid power needed	Cuts spoilage sharply, works off grid, no equipment cost to farmers	Not a marketplace, does not solve transport, physical scaling is slow
AFEX	Warehouse receipts as loan collateral, blockchain verification, 45+ warehouses	Unlocks working capital, lowers lender risk, proven fundraising track record	Built for durable grains not perishables, needs formal warehouse access
Complete Farmer	Three sided app, blockchain traceability, cold chain packhouses	Cuts out middlemen, strong buyer trust, grew to 50,000+ farmers	Fulfilment centres are the real bottleneck, mostly export focused
Twiga Foods	Owned farms, trucks and warehouses, mobile ordering app	Massive scale at peak, guaranteed quality from owned tomato farms	Vertical ownership proved too costly, forced 2023-2025 restructuring
ThriveAgric	Input financing, farm data platform, guaranteed offtake deals	500,000+ farmers financed, yields up to 3x national average	Grain focused, does not address storage or transport directly
Farmcrowdy	Public crowdfunding of individual farms, sponsor updates	Opened farm investment to the public, raised about \$15M at peak	Exited crowdfunding under regulatory pressure, now inactive
e-Choupal	Shared village internet kiosks, direct sale to ITC	Removed intermediaries at national scale, proven over two decades	Fixed shared kiosks, slow to modernise, built for durable crops



Each platform above addresses a single link, storage, financing, trading or logistics, in isolation. TomaLink brings storage, transportation, a verified marketplace and live tracking into one connected platform, so a farmer never has to leave the app to solve the next problem in the chain.



ABOUT US

TomaLink is a technology driven post harvest solution that connects farmers to cold storage, warehousing, logistics and markets. By reducing post harvest losses, TomaLink helps farmers earn more, minimises food waste, and strengthens Nigeria's tomato value chain.

TAGLINE

From Farm to Market, Fresh Every Step.

OUR VALUES

Innovation

Solving old problems with new, practical technology

Reliability

A platform farmers and buyers can depend on every season

Sustainability

Reducing waste across the whole value chain, not just one link

Farmer Empowerment

Putting pricing power and market access back in farmers' hands

Transparency

Clear pricing, verified buyers, visible transactions

Collaboration

Every stakeholder working from the same shared platform

OUR BRAND

A tomato and a chain, the two ideas at the heart of TomaLink

Logo Presentation

ii

7/2026

Logo Rational

The TomaLink logo combines two core elements: the tomato and the chain.

The tomato represents agriculture, freshness, and the harvest, while the chain symbolizes connection across the tomato value chain—from farmers to consumers through technology and efficient logistics.

The wordmark uses Sora, a modern geometric sans-serif chosen for its clean, bold, and approachable style, reflecting TomaLink's values of innovation, reliability, and accessibility.

Together, these elements communicate TomaLink's mission of preserving every harvest through connection and innovation.



OUR APPROACH

One platform, four connected services, a single farmer journey

Store

Book verified cold storage and warehouse space near the farm the moment harvest happens, instead of leaving produce exposed.

Move

Schedule refrigerated transport with transparent pricing, so tomatoes travel cold instead of bruised and overheated.

Sell

List produce directly to verified buyers, wholesalers, retailers and processors, with live market prices guiding every decision.

Track

Follow every shipment from pickup to delivery in real time, with proof of delivery and ratings closing the loop.

Existing solutions only address one part of the supply chain. TomaLink integrates storage, transportation, marketplace and tracking into one seamless platform.

OBJECTIVES

What success looks like, for the business and for the people using it

BUSINESS OBJECTIVES

- **Reduce post harvest losses by 30% within the first year**
- **Increase average farmer income by 20% within 12 months**
- **Cut the average farm to market time by 25%**
- **Onboard 5,000 farmers, 1,000 buyers, 100 logistics providers and 50 storage partners in year one**
- **Build sustainable revenue through commissions on bookings and sales**

USER OBJECTIVES

- **Give farmers and traders easy access to nearby cold storage**
- **Let users search, compare and book storage and transport in one place**
- **Connect farmers directly with verified buyers, cutting out unnecessary middlemen**
- **Show live prices, storage availability and delivery status at all times**
- **Keep the whole experience simple, transparent and fast to use**

WHO USES TOMALINK

Five roles, one coordinated platform

Farmer

**Registers harvests,
lists produce,
requests transport,
books storage,
tracks sales and
receives payment.**

Buyer

**Searches available
produce, places
orders, pays
securely, tracks
delivery and rates
farmers.**

Transporter

**Accepts requests,
schedules pickups,
updates delivery
status and confirms
completed deliveries.**

Warehouse Owner

**Lists storage
capacity, accepts
bookings, monitors
stored produce and
flags nearing expiry.**

Administrator

**Verifies accounts,
approves partners,
resolves disputes
and oversees the
whole platform.**

CORE FEATURES

The ten must have features that define the MVP

01 User Registration & Authentication

03 Cold Storage Booking

05 Logistics Booking

07 Order Management

09 Payment Integration

02 User Profile Management

04 Warehouse Booking

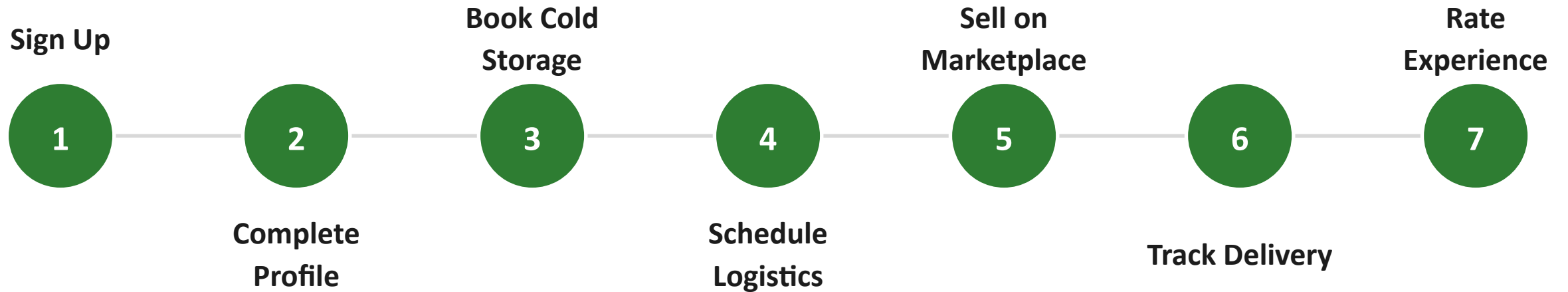
06 Marketplace

08 Shipment Tracking

10 Admin Dashboard

THE FARMER JOURNEY

From sign up to a successful sale, in seven simple steps



Each step closes a gap where produce is normally lost. Booking storage the moment harvest happens prevents open air spoilage, scheduling refrigerated transport prevents bruising in transit, and live tracking means a shipment is never simply lost to silence.

INSIDE THE PRODUCT

Getting started takes under a minute

Log in/Sign up


FARMER


STORAGE FACILITATORS


BUYERS

Choose a role, Farmer, Storage Facilitator or Buyer



WELCOME BACK!

Email or phone

Password

[Forgot Password?](#)

Log in

Create account

or continue with

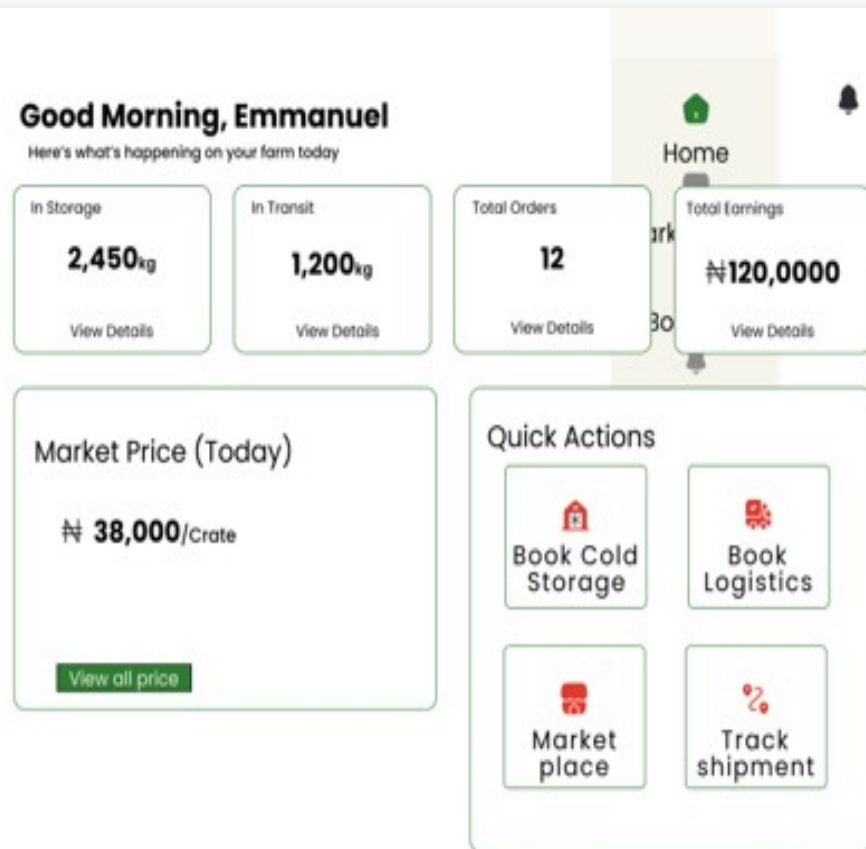


Continue with Google

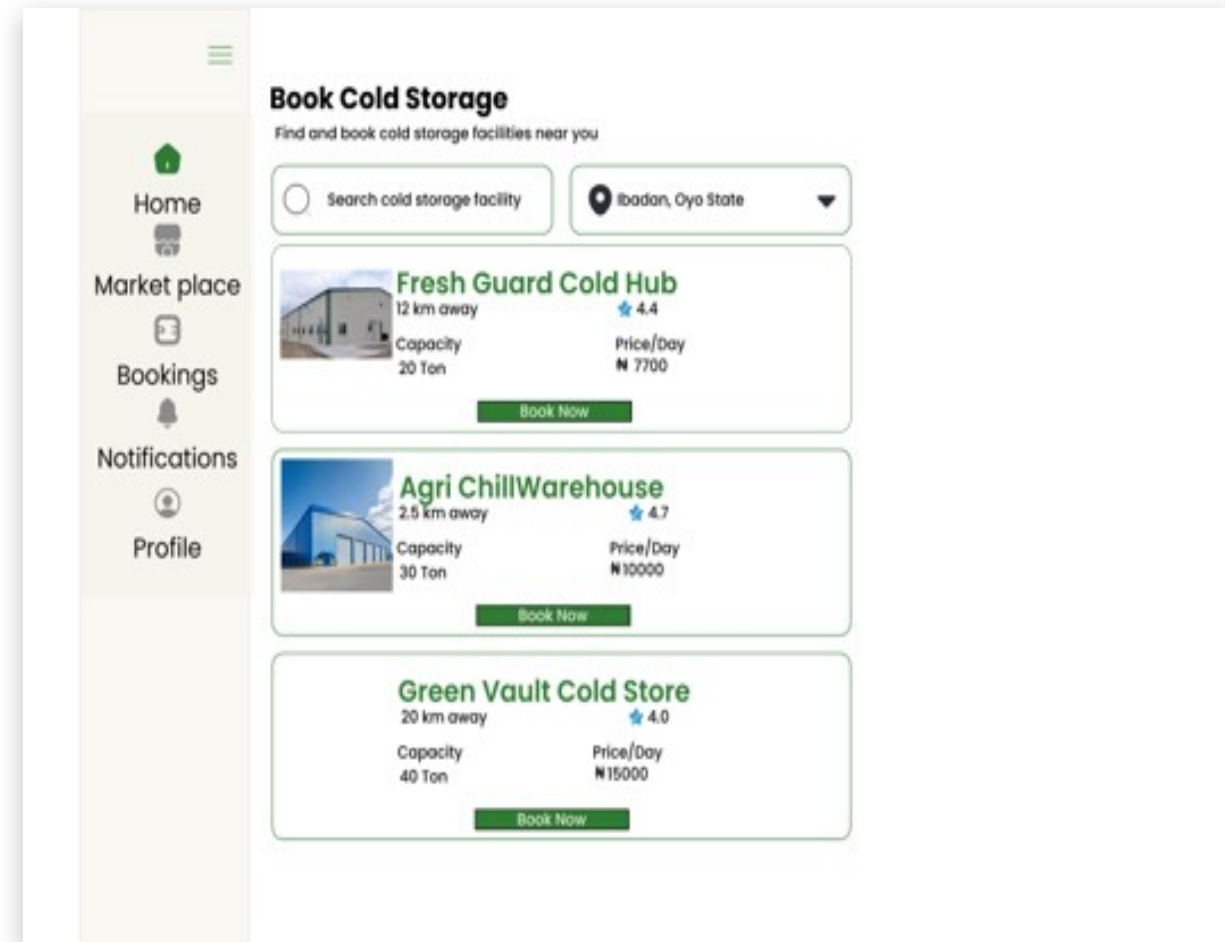
Sign in with phone, email or Google

3.0 INSIDE THE PRODUCT

One dashboard, then storage booked in a few taps



A single home screen for storage, logistics, market and sales



Compare nearby cold storage by price, capacity and distance

INSIDE THE PRODUCT

Book a refrigerated truck, then watch it move in real time

The screenshot shows the 'Book Logistics' form in a mobile application. On the left is a vertical sidebar with icons and labels for 'Home', 'Market place', 'Bookings', 'Notifications', and 'Profile'. The main content area is titled 'Book Logistics' with the subtitle 'Find and book logistics near you'. It contains several input fields: 'Pickup Location' (Farmgate, Ibadan, Oyo State), 'Destination' (Oyingbo Market, Lagos State), 'Tomato Quantity (kg)' (2000kg), and 'Preferred Pickup Date' (27 July, 2026). Below these is a 'Vehicle Type' section with 'Refrigerated Truck 10 Ton' selected, and an 'Estimated Price' of '₦120,000'. A green button at the bottom is labeled 'Request Refrigerated Truck'.

Request a refrigerated truck with a clear price upfront

The screenshot shows the 'Track Shipment' page in a mobile application. The sidebar is identical to the previous screen. The main content area is titled 'Track Shipment' with the subtitle 'Track the location of your order'. It features a driver profile for 'Abiola David' with a 4.6 rating and a map showing the truck's route from 'Oyingbo Market, Lagos State' to 'Farmgate, Ibadan, Oyo State'. Below the map is a 'Live Status' section with a timeline: 'Pickup Confirmed' (20 July, 2026, 10:30), 'In Transit' (20 July, 2026, 02:15), 'Arriving' (20 July, 2026, 05:30), and 'Delivered' (Pending). On the right, there are three green buttons: 'Call Driver', 'Chat', and 'Delivery Receipt'.

Live GPS tracking from pickup confirmed to delivered

HOW WE BUILT IT

A modern stack, secured by design, built with AI as a reviewed co-pilot

TECHNOLOGY STACK

- Frontend: HTML, CSS, JavaScript, React
- Backend: Node.js, Express
- Database: MongoDB
- Workflow: Jira, GitHub, Figma, Canva, Google Workspace

SECURITY HIGHLIGHTS

- Secure authentication and password protection
- Role based access for each of the five user types
- Data encryption and privacy protection throughout
- Active threat mitigation on all user facing endpoints

Date	Team	AI Tool	Purpose	AI Output	Human Review / Validation
15 Jul	Product Management	ChatGPT	PRD, user stories	PRD draft	Reviewed by team
15 Jul	Virtual Assitant	ChatGPT	Meetings/Jir a	Minutes & plans	Reviewed
16 Jul	UI/UX	ChatGPT	Layouts & UX	Design ideas	Refined
16 Jul	Frontend	ChatGPT	React/debug ging	Code snippets	Tested
16 Jul	Backend	ChatGPT	API/backend	Architecture ideas	Validated
17 Jul	Data Analysis	ChatGPT	Research/in sights	Analysis	Verified
17 Jul	Data Science	ChatGPT	Algorithms	ML guidance	Reviewed
17 Jul	Brand Design	ChatGPT	Brand ideas	Logo concepts	Refined
18 Jul	Digital Marketing	ChatGPT, Gamma AI	Marketing	Campaign ideas	Reviewed
18 Jul	Cybersecurit y	ChatGPT	Security	Recommend ations	Validated 
18 Jul	AI & ML	ChatGPT	Models/doc umentation	Model outputs	Tested & edited

Every AI assisted output across product, design, engineering and marketing was reviewed, tested or validated by a team member before use, logged here for full transparency.

ROADMAP & GO TO MARKET

Six months from research to a national launch



GO TO MARKET, THREE PHASES

Awareness

4 weeks

Landing page, social campaigns, farming community outreach. Target: 2,000+ waitlist signups.

Pilot Launch

2 weeks

Live In Kaduna, Kano, Plateau and Jos with free first bookings. Target: 500 active users, 100 bookings.

Public Launch

Ongoing

Launch event, press coverage, referral programme and NGO partnerships. Target: 5,000 users in year one.

SUCCESS METRICS

How we will know TomaLink is working

6,000+

registered users within Year 1

≥ 85%

booking completion rate

₦50M+

annual transaction value

≥ 60

Net Promoter Score

30%

reduction in post harvest losses

20%

increase in average farmer income

EXPECTED IMPACT

Better outcomes for every stakeholder in the chain

Farmers

Higher income, lower spoilage,
direct access to buyers

Businesses

Efficient logistics, reliable supply,
improved produce quality

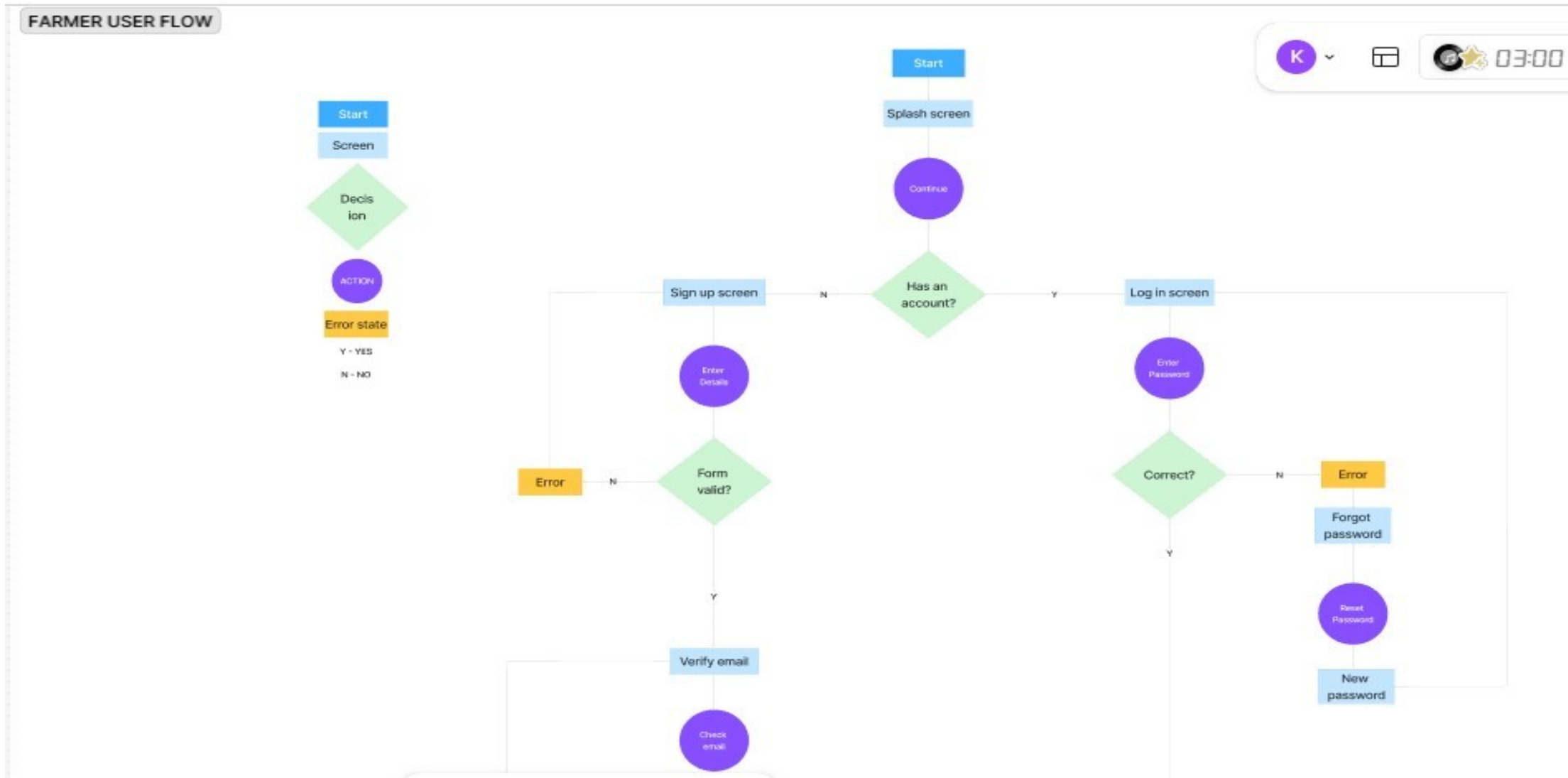
Nigeria

Reduced food waste, stronger food
security, digital agriculture adoption



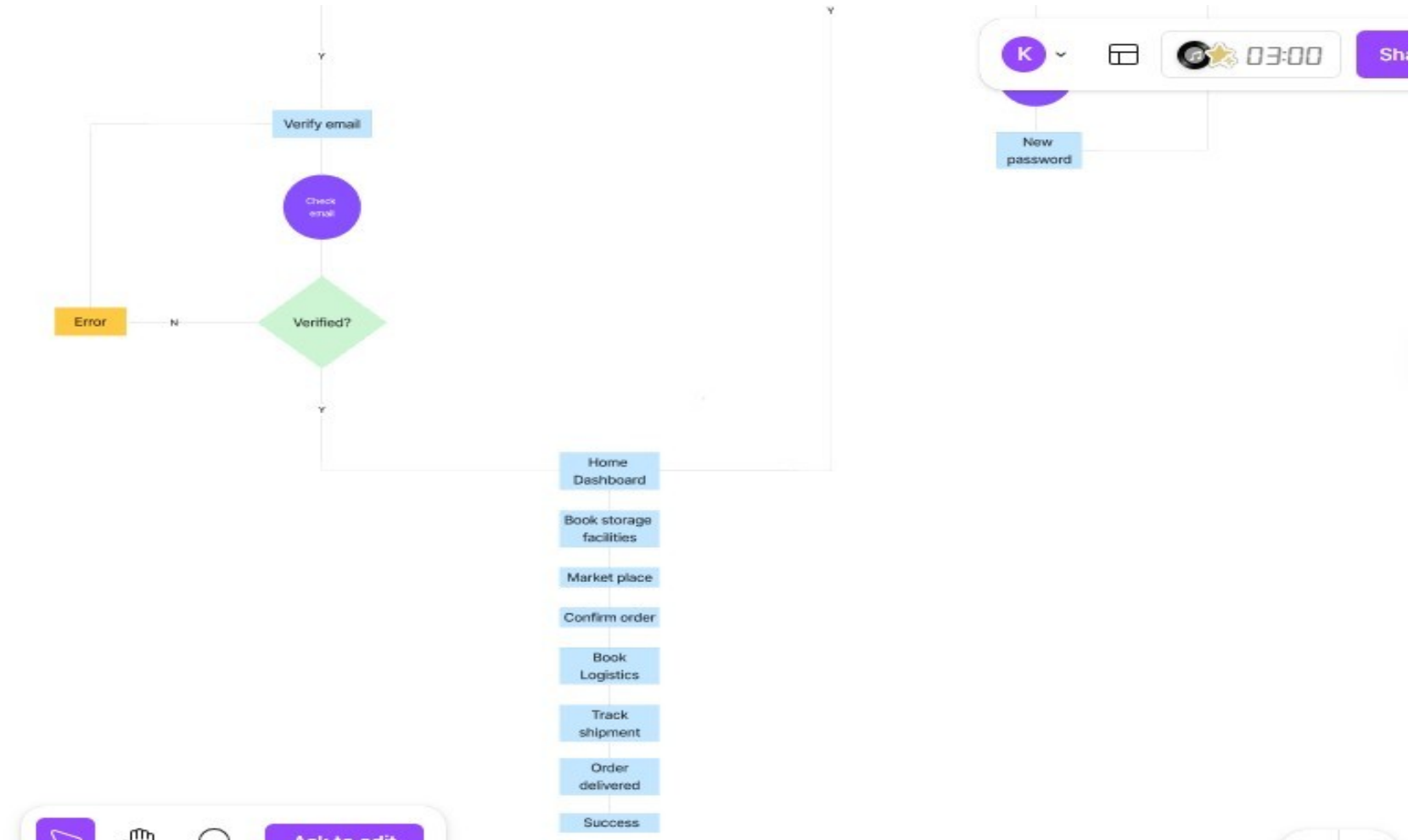
UI/UX, FARMER USER FLOW

Onboarding and sign in, before a farmer ever reaches the dashboard



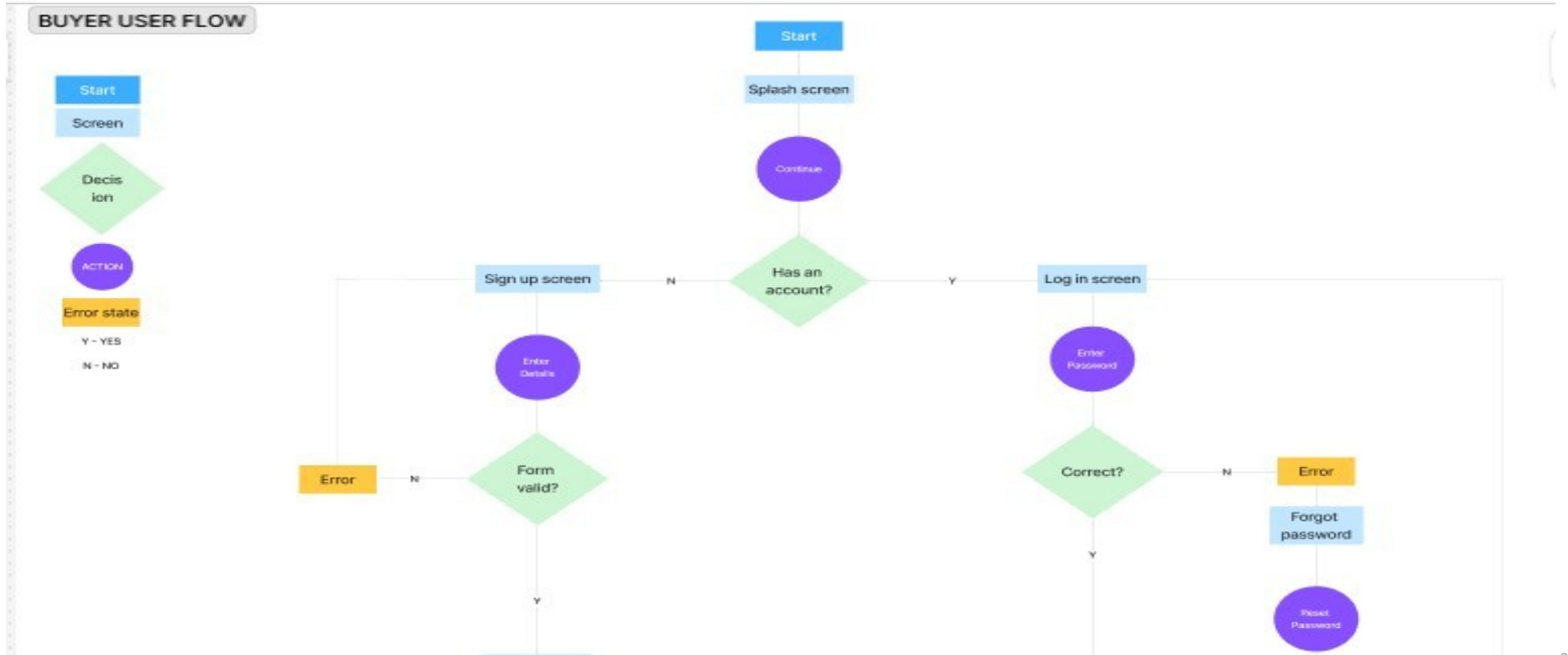
UI/UX, FARMER USER FLOW

From email verification through to a listed harvest



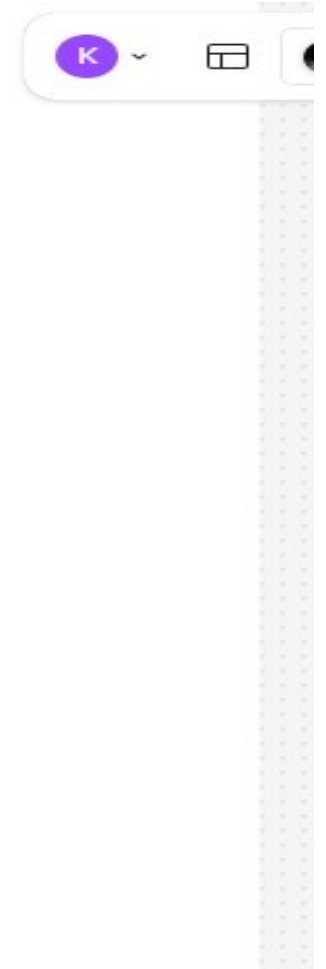
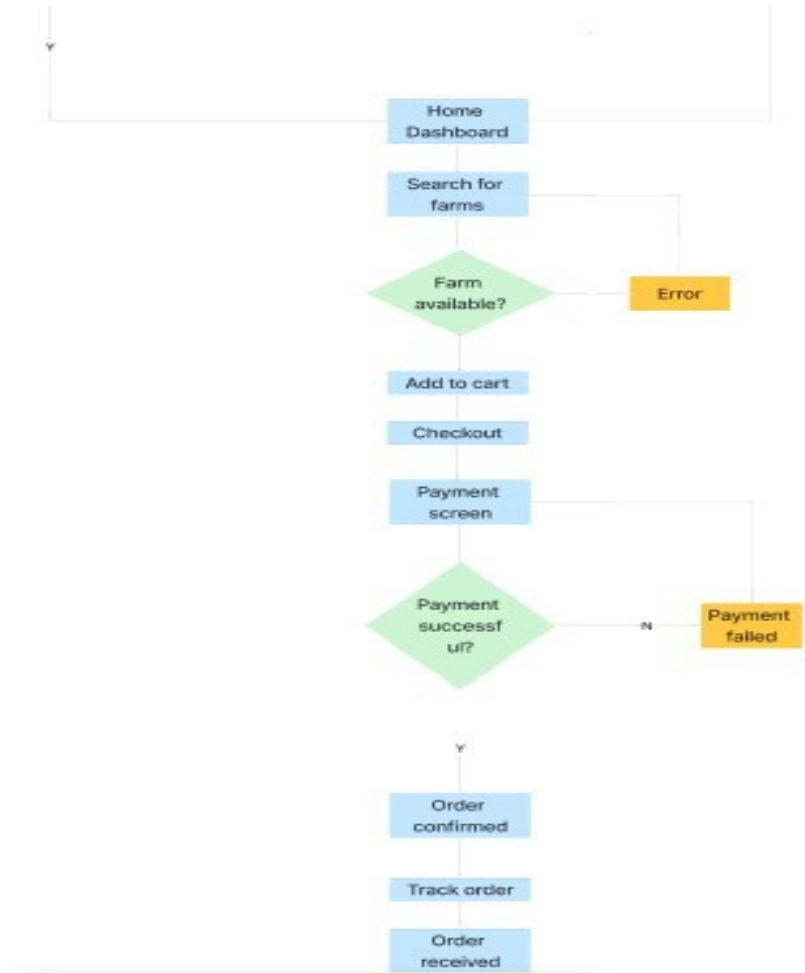
UI/UX, BUYER USER FLOW

The same trusted onboarding pattern, built for buyers

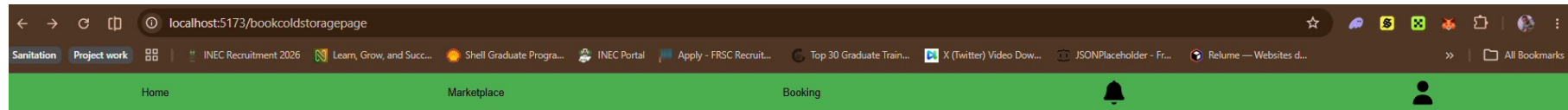


UI/UX, BUYER USER FLOW

From browsing the marketplace through to a received order



LIVE PRESENTATION



Book Cold Storage

Find and book cold storage facilities near you



Fresh Guard Cold Hub

12km away
Capacity
20 Ton

★ 4.4
Price/Day
₦7,700

Book Now



Agri ChillWarehouse

2.5km away
Capacity
20 Ton

★ 4.7
Price/Day
₦10,000

Book Now



Green Vault Cold Store

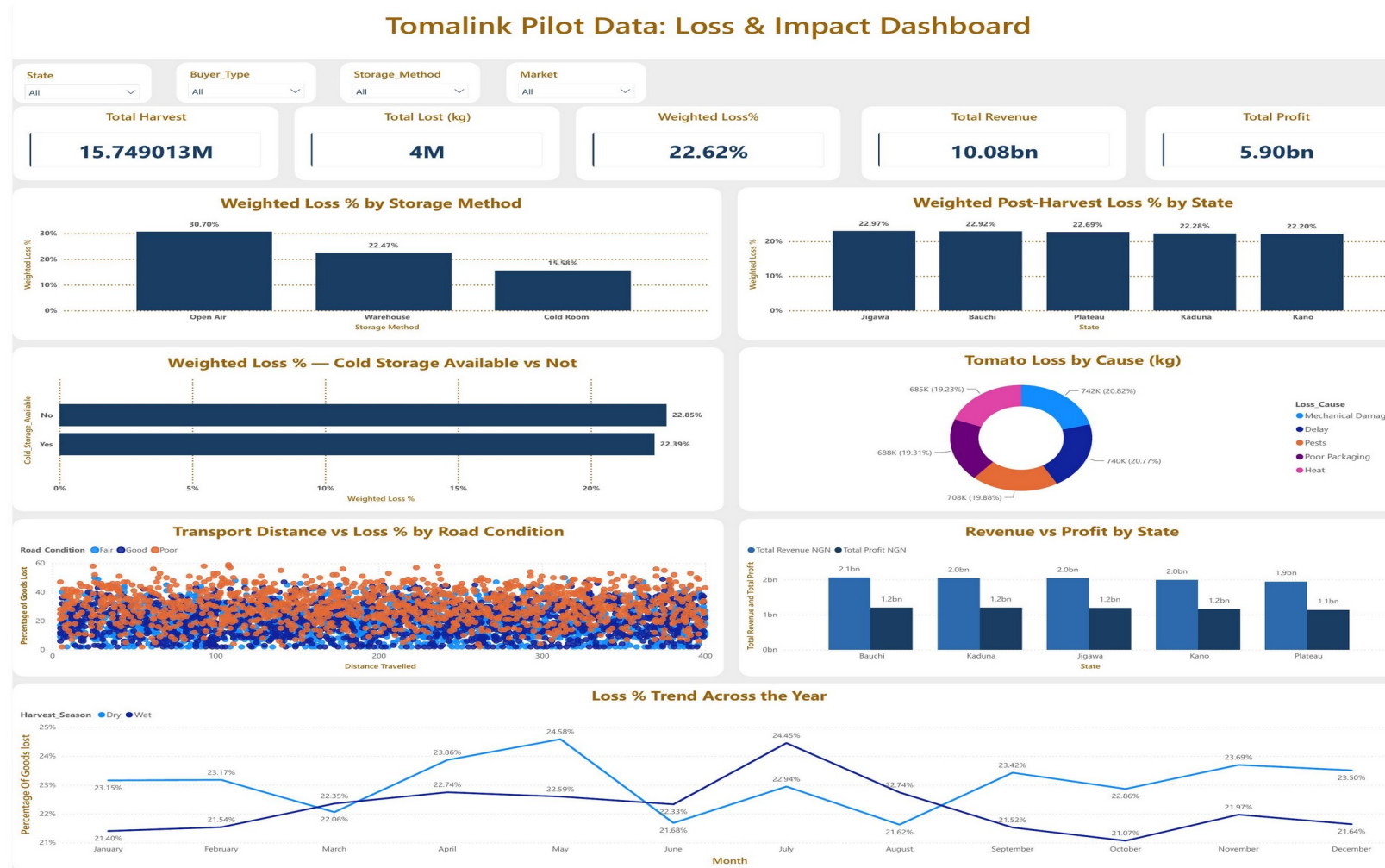
20km away
Capacity
20 Ton

★ 4.0
Price/Day
₦15,000

Book Now

DATA ANALYSIS

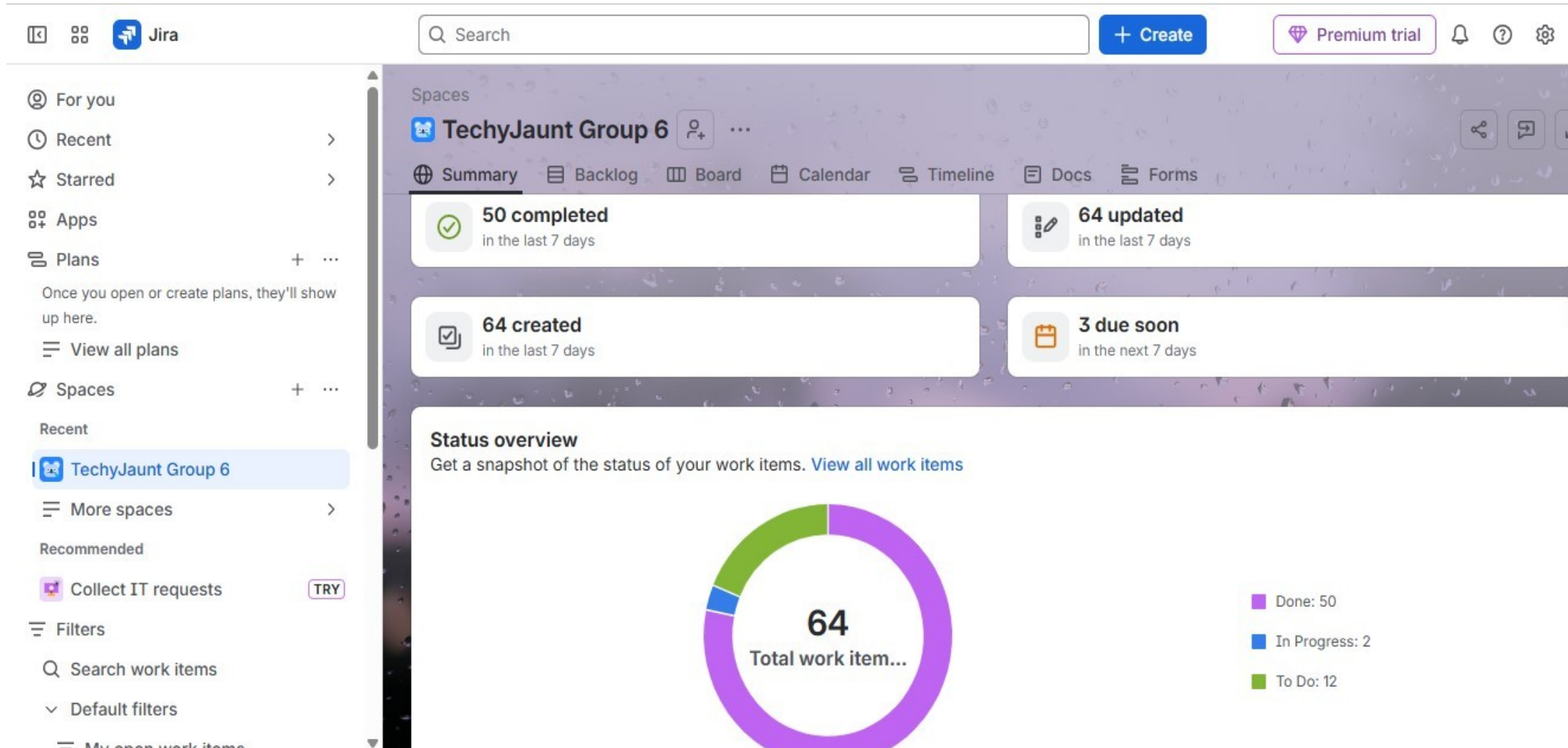
One pilot dashboard, turning raw loss data into a clear story



Built from real pilot records across five tomato producing states, covering harvest volume, loss causes, storage method, transport conditions and profitability.

VIRTUAL ASSISTANT, PROJECT COORDINATION

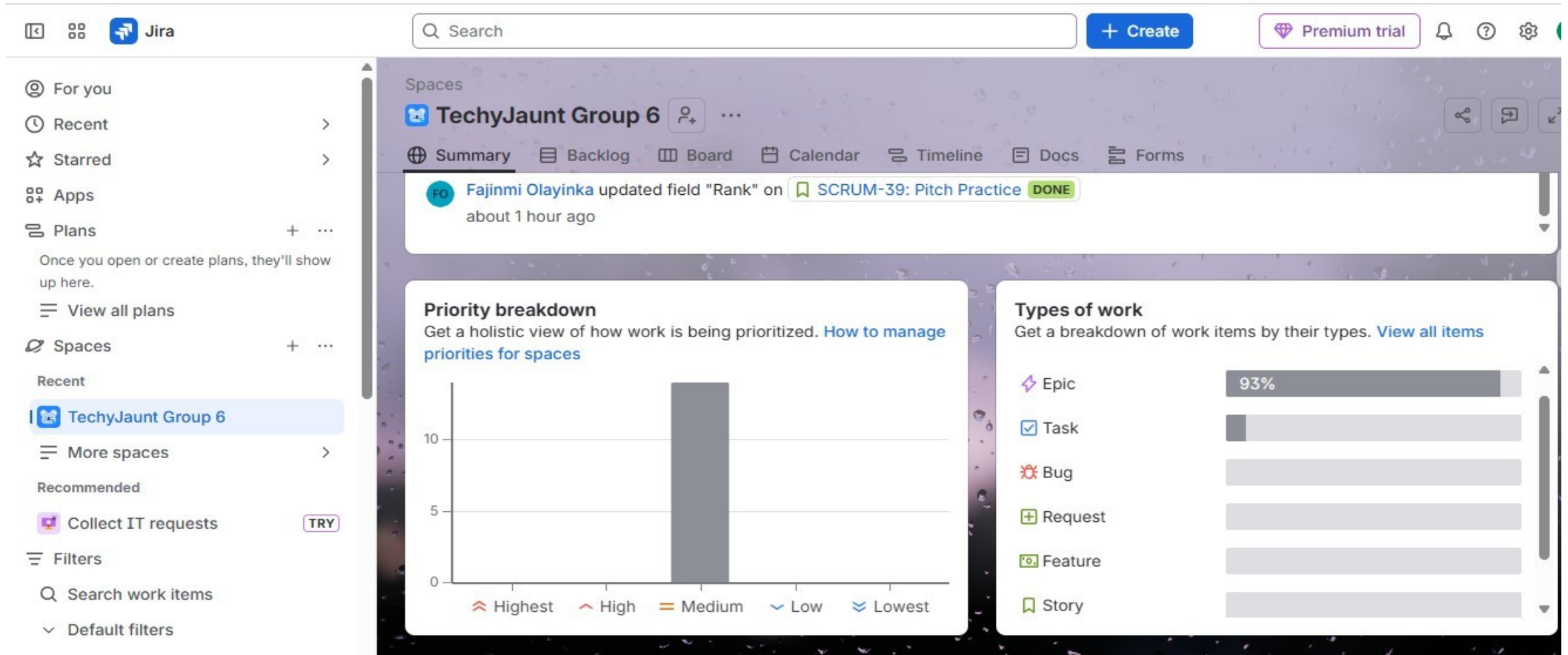
Sprint activity tracked and coordinated end to end in Jira



50 tasks completed and 64 updated in the last 7 days, with 64 total work items tracked across the team.

VIRTUAL ASSISTANT, PROJECT COORDINATION

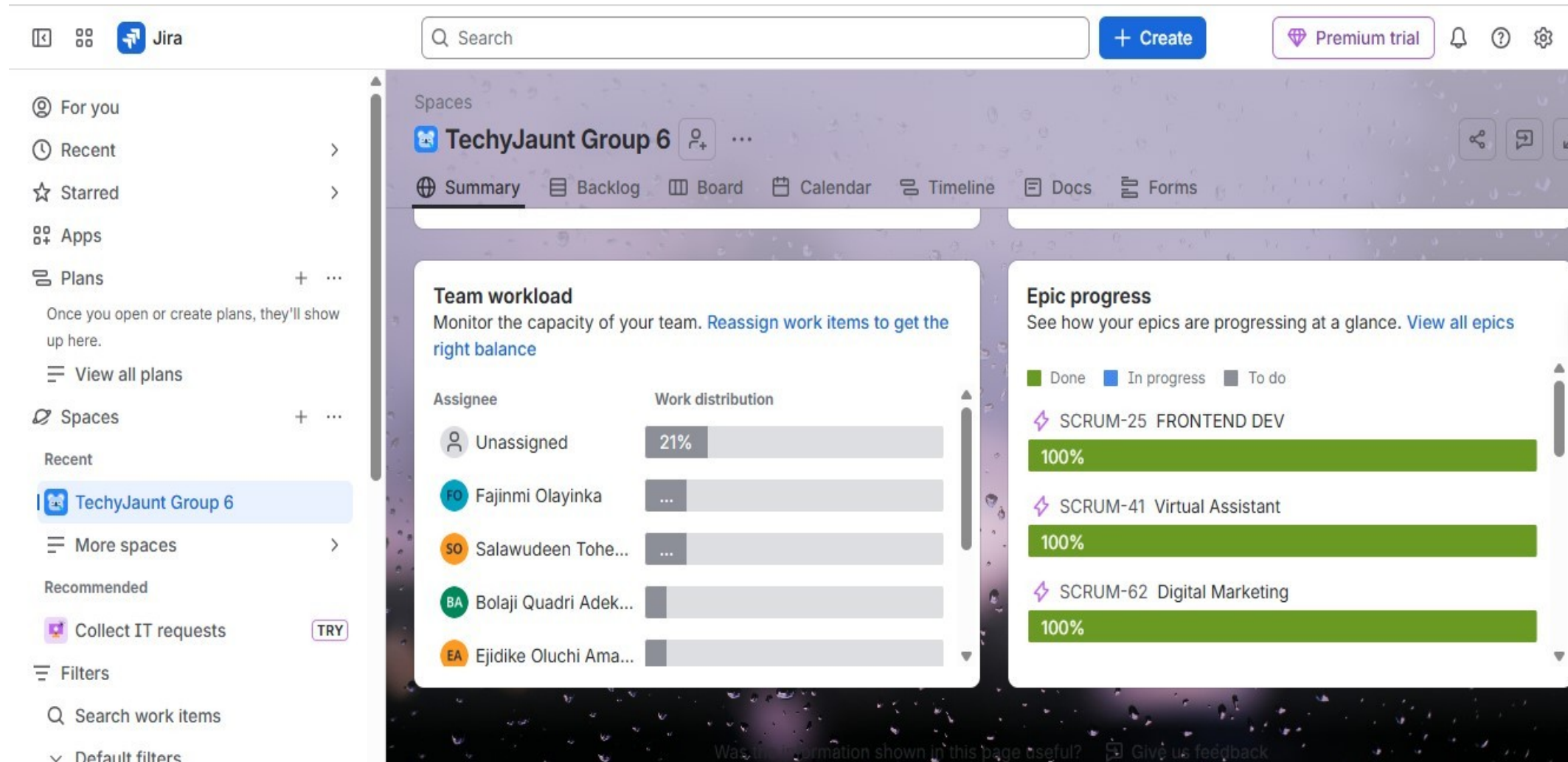
Work was prioritised and typed for full visibility across teams



Nearly all work items were tagged Medium priority, and the majority were tracked as Epics spanning each department's larger deliverables.

VIRTUAL ASSISTANT, PROJECT COORDINATION

Every department's epic closed out at full completion



Frontend Development, Virtual Assistant and Digital Marketing epics each reached 100% completion, with workload monitored and rebalanced across the team throughout.

PRODUCT MANAGERS

Salawudeen Toheeb Olalekan
Elizabeth Oluwabusola Sonola
Vanessa Adetoro osarobo
Michael Alao

VIRTUAL ASSISTANTS

Akinpade Jesudasimi
Fajinmi Olayinka
Olawunmi Funmilola
Esther Orji
Norah Egwakhide
Precious Temidayo

FRONTEND DEVELOPER

Osazuwa Kelvin
Olayiwola Rakanmi

CONTRIBUTION

- Product planning
- Sprint management
- Requirements gathering
- Team coordination

CONTRIBUTION

- Meeting scheduling
- Documentation
- Reports
- Team communication
- Sprint coordination
- Project tracking

CONTRIBUTION

- Developed the application's user interface
- Connected screens • Improved user experience

BACKEND DEVELOPER

Oluchi Ejidike

CONTRIBUTION

- Built APIs
- Authentication
- Database connection
- Business logic

DATA SCIENTISTS

Hassan Mariam

CONTRIBUTION

- Dataset preparation
- Data cleaning
- Exploratory Data Analysis
- Feature engineering
- Prepared data for Machine Learning

UI/UX DESIGNERS

Oni Samuel
Ndubuisi Victor

CONTRIBUTION

- User research
- Wireframes
- High-fidelity designs
- Prototype

CYBERSECURITY

Bolaji Quadri Reign-El Alajemba

CONTRIBUTION

- Risk Assessment
- Authentication
- Data Privacy
- Security Recommendation

DIGITAL MARKETING

Atokoh Naomi Nwaduito
Ademola Raifu
Angela Silas

CONTRIBUTION

- Branding
- Marketing Strategy
- Go-To-Market Plan

AI & ML MODULE

Aminullahi Zainab Olamide

CONTRIBUTION

- Built predictive model
- Tested ML algorithms
- Generated spoilage predictions

DATA ANALYSTS

Suleiman Titilayo Khadijat
Eunice Istifanus Ishaku
Ulasi Chukwuma Obinna
Ogwu Joel Edike
Chiziterem Ugorji
Orija Rasheedah

CONTRIBUTION

- Research
- Market analysis
- Dashboard
- Insights
- Validation of solution

**THANK
YOU
FOR
LISTENING**

